



W1.2 COS30045

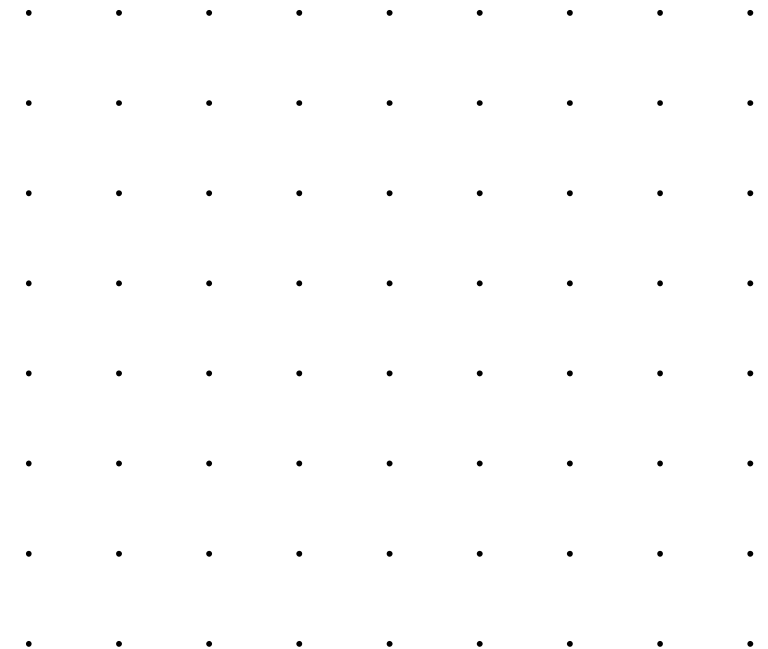
Lecture

Agenda

- Data attributes and statistics
- Data quality
- Data preparation
 - Identifying issues
 - Fixing issues
 - Data transformation

References

Han, Kamber, Pei (2011) *Data Mining Concepts*
Ch 2.1, 2.2
Ch 3.1, 3.2, 3.5



Data attributes and descriptive statistics

Data attributes will determine how we:

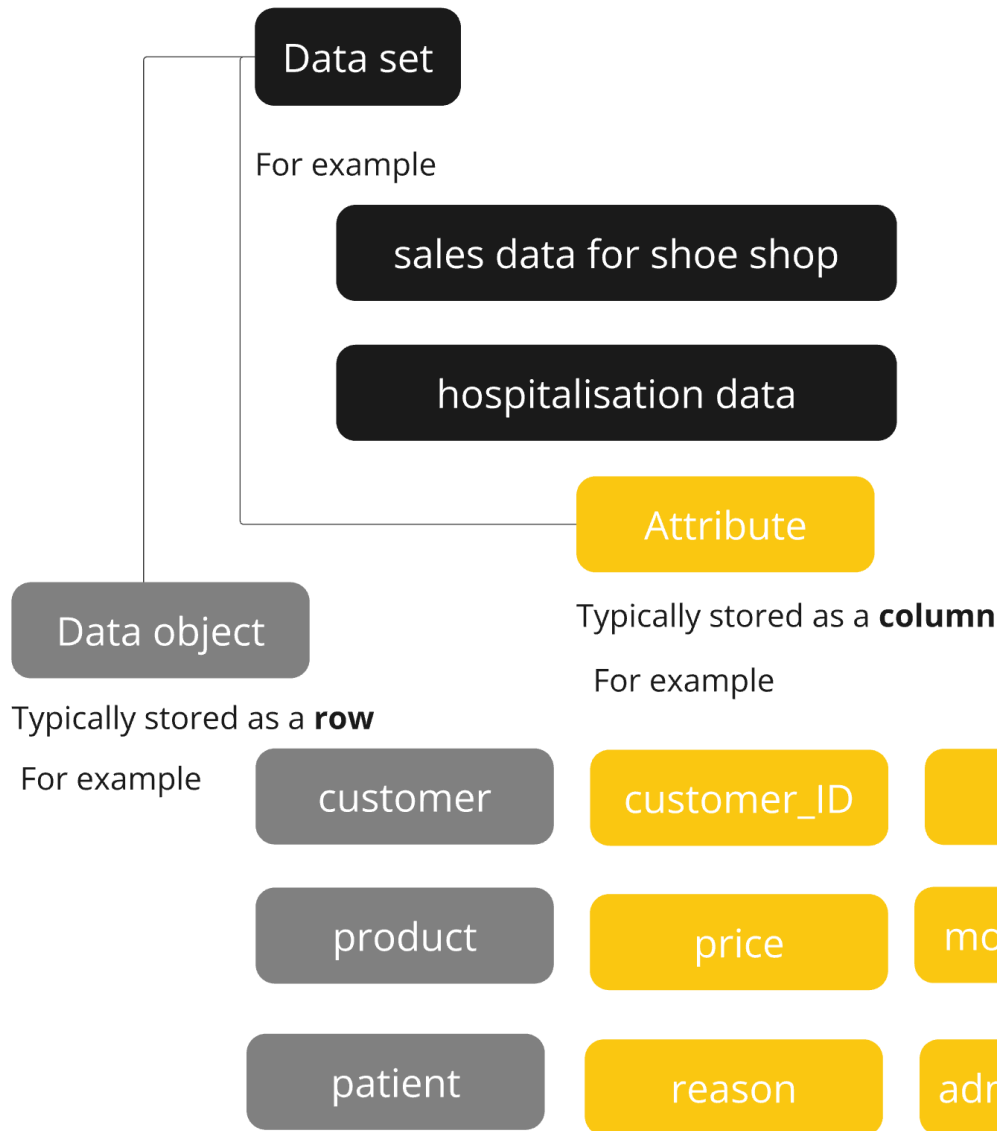
- Measure central tendency (middle value)
- Clean the data
- Transform the data
- Visualise the data

Descriptive statistics will help us:

- Work out what to do about missing values
- Spot outliers



Naming conventions

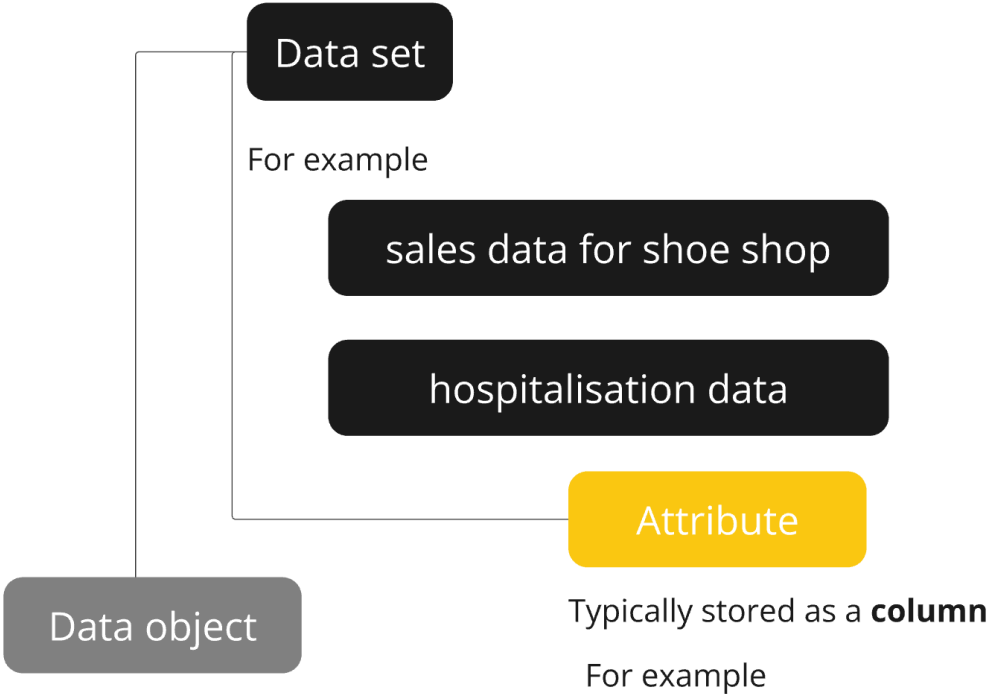


Data set: Tasmanian Wombat Population Survey

Rows: 19548 | Columns: 8

	#	RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)
☐	1	Row0	1	28	0	5	Ansons-1	2002	Modified land	0
☐	2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
☐	3	Row2	3	28	10	20	Ansons-1	2002	Modified land	0
☐	4	Row3	4	28	20	40	Ansons-1	2002	Modified land	0
☐	5	Row4	5	28	40	60	Ansons-1	2002	Modified land	0
☐	6	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
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☐	8	Row7	8	29	5	10	Ansons-1	2003	Modified land	0
☐	9	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
☐	10	Row9	10	29	20	40	Ansons-1	2003	Modified land	0

Naming conventions



Typically stored as a **row**
For example

customer	customer_ID	address	items purchased	.	.	.	.	.	.
product	price	model number	items sold	.	.	.	.	.	.
patient	reason	admission date	days in care	.	.	.	.	.	.



Data set: Tasmanian Wombat Population Survey

Rows: 19548 Columns: 8		Table Statistics							
#	RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)
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Tuple

Attribute (aka dimension, feature, , variable)

An *attribute* is a data field, representing a characteristic or feature of a data object.

Han, Kamber, Pei (2011)

Data set: Tasmanian Wombat Population Survey

Data objects

Attributes

Rows: 19548 | Columns: 8

Table

Statistics

☐	#	RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)	☐
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☐	4	Row3	4	28	?	40	Ansons-1	2002	Modified land	0	
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☐	7	Row6	7	29	0	5	Ansons-1	2003	Modified land	0	
☐	8	Row7	8	29	5	10	Ansons-1	?	Modified land	0	
☐	9	Row8	9	29	10	20	Ansons-1	2003	Modified land	0	
☐	10	Row9	10	29	20	40	Ansons-1	2003	Modified land	0	

Attribute

Attribute Type

Nominal: name (category, code)

Binary: 0 or 1

Ordinal: ranking order

Numeric: quantitative, measurable, integer

- Interval (equal size units)
- Ratio (measurement with zero)



Attribute

- Attribute: **Survey_ID**
 - Observed value: 28
 - Type: nominal
- Attribute: **Dist_Min**
 - Observed value: 0
 - Type: ratio
- Attribute: **Year**
 - Observed value: 2002
 - Type: interval

Data set: Tasmanian Wombat Population Survey

Rows: 19548 | Columns: 8

Table Statistics

#	RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)
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3	Row2	3	28	10	20	?	2002	Modified land	0
4	Row3	4	28	?	40	Ansons-1	2002	Modified land	0
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6	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
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Measures of central tendency

The 'centre' of the data set (i.e., where most values fall)

Mean

- sum of values $\div$ number of values
- sensitive to outliers (extreme values)

Median

- Middle value of ordered set
- Better for skewed data

Mode

- Most frequent
- Good for nominal data

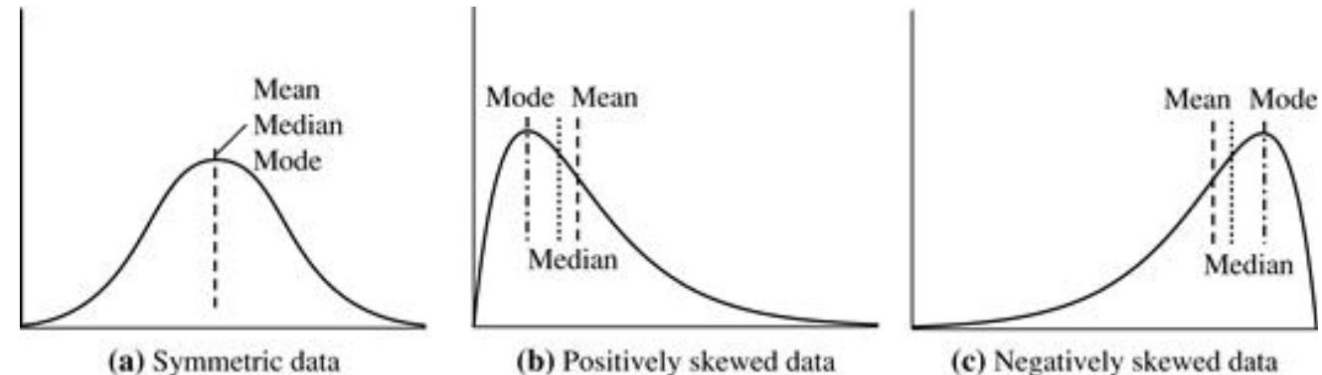


FIGURE 2.1 Mean, median, and mode of symmetric versus positive- and negatively skewed data.

Data Quality

- Accuracy
- Completeness
- Consistency
- Timeliness
- Believability
- Interpretability



Data Quality

- Accuracy
- Completeness
- Consistency

Some sources of quality issues

- Faulty instruments
- Human/computer error
- Naming conventions
- Data in different formats (cm/in) etc

tv_2025_01_21.csv

Model no.
QA50LS03TAW

ScreenSize
125.725 cm -> 49 inches

Pasv_stn... String	Act_stnd_power String	Act_stnd_time String
0.46	1.26	10.0
0.401	0.401	7.0
0.401	0.401	7.0
0.25	0.25	7.0
-	-	-
0.4	0.0	0.0
0.4	0.0	0.0
0.4	0.0	0.0
0.4	0.0	0.0
0.45	0.46	7.0
-	-	-
-	-	-

Data Quality

- Timeliness
 - Data not updated regularly/in time for reports
- Believability
 - User trust



Data Quality

- Interpretability
 - Data dictionary
 - Understanding transformations



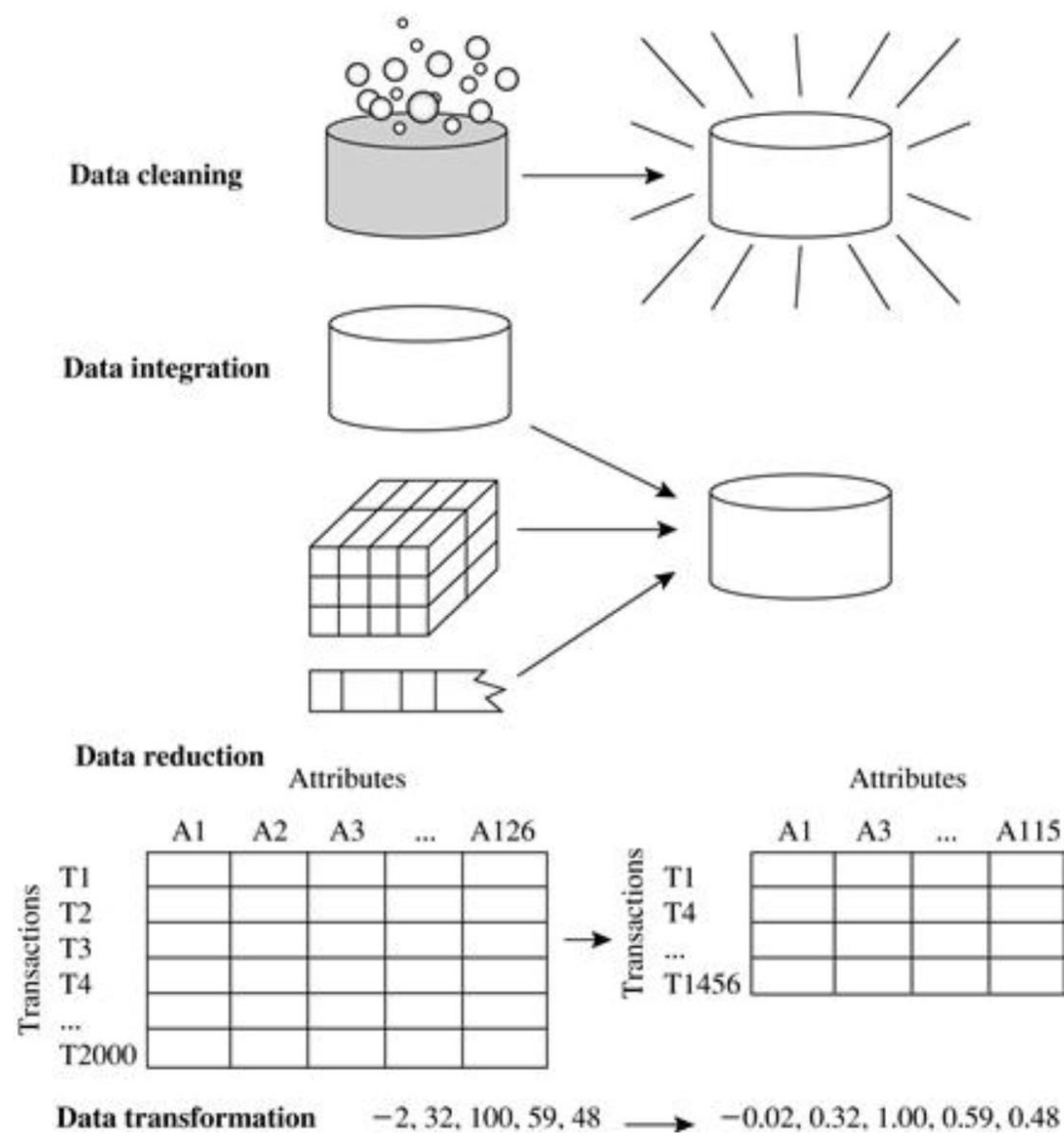
DATA-SPECIFIC INFORMATION FOR: MAINLAND_HABITAT.csv

1. Short description: shows various data relating to Tasmanian mainland spotlight surveys
2. Number of variables: 11
3. Number of cases/rows: 19548
4. Variable list:
 - Surveys.Survey_ID: individual ID assigned to each individual spotlight survey
 - Dist_Min: minimum distance (in metres) of wombat sightings/counts according to previously set distance intervals/bands
 - Dist_Max: maximum distance (in metres) of wombat sightings/counts according to previously set distance intervals/bands
 - Route_ID: individual ID assigned to each spotlight survey transect
 - Year: year of the spotlight survey (2002-2020)
 - Veg_Group: describes main vegetation group the transect runs through; 11 vegetation groups as defined by Kitchener, A. & Harris, S. (2013). From Forest to Fjaeldmark: Descriptions of Tasmania's Vegetation. Australia: Department of Natural Resources and Environment, Tasmania and used in the TASVEG 4.0 dataset
 - Sightings: number of wombats counted
 - DistBand: For each sighted/counted wombat, perpendicular distance from transect was recorded, using six categories/bands: 0-5, 5-10, 10-20, 20-40, 40-60, and 60-100 m (metres).
 - j,k,i: numerical classifications for 'vegetation group', 'year' and 'distance band/interval', respectively
5. Specialised formats or other abbreviations used: None

Data Preprocessing

- Data cleaning
- ~~Data reduction~~
- Data transformation
- Data integration

Garbage in, garbage out.



Data Cleaning: Missing Values

Ignore the tuple

- Delete row from data set
- Can reduce data set dramatically for only one missing attribute

Fill in the missing value manually

- Time consuming for large data sets

Use global constant

- Replace with 'unknown', 'NA'
- Take care how this might be used in analysis



Data Cleaning: Missing Values

Use measure of central tendency based on:

- Mean of entire data set
- Mean of data objects within same category
- Median (if data skewed)

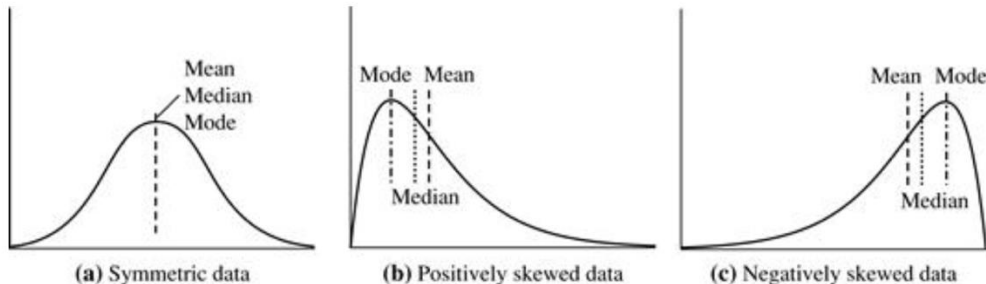
Example:

Data set: Basketball player stats

Issue: Missing height data for player

Solution: Replace with data set mean?

- Not if male and female players in data set!



Missing Data

Imputation

- Can use attribute of belonging to same class

Mean (Download Speed) = 130

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	157	80%
2	Lite	99	70%
3	Fast+	167	10%
4	Fast+	N/A	80%
5	Lite	76	70%
6	Fast+	155	10%
7	Fast+	N/A	95%
8	Lite	76	77%
9	Fast+	180	95%



Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	157	80%
2	Lite	99	70%
3	Fast+	167	10%
4	Fast+	130	80%
5	Lite	76	70%
6	Fast+	155	10%
7	Fast+	130	95%
8	Lite	76	77%
9	Fast+	180	95%

Median (Download Speed) = 155

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	157	80%
2	Lite	99	70%
3	Fast+	167	10%
4	Fast+	N/A	80%
5	Lite	76	70%
6	Fast+	155	10%
7	Fast+	N/A	95%
8	Lite	76	77%
9	Fast+	180	95%



Mobile ID	Mobile Package	Download Speed	Data Limit Usage
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3	Fast+	167	10%
4	Fast+	155	80%
5	Lite	76	70%
6	Fast+	155	10%
7	Fast+	155	95%
8	Lite	76	77%
9	Fast+	180	95%

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
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5	Lite	76	70%
6	Fast+	155	10%
7	Fast+	N/A	95%
8	Lite	76	77%
9	Fast+	180	95%

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	157	80%
3	Fast+	167	10%
4	Fast+	N/A	80%
6	Fast+	155	10%
7	Fast+	N/A	95%
9	Fast+	180	95%

Mean Fast+ = 165
Replace the N/A with 165 in this example.

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
2	Lite	99	70%
5	Lite	76	70%
8	Lite	76	77%

Identifying issues

Check data dictionary to understand data

Detecting discrepancies

- Poorly designed data collection
- Human error
- Deliberate error
- Outdated data
- Inconsistent use of codes

Find anomalies

- Do some basic stats:
 - skewed



Identifying issues

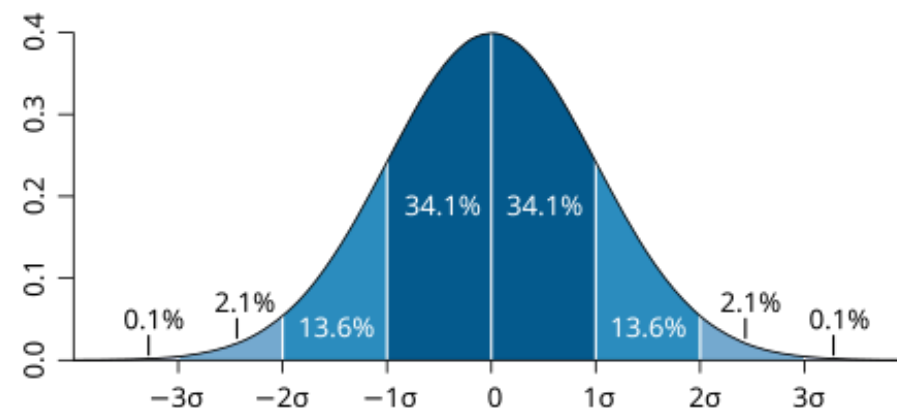
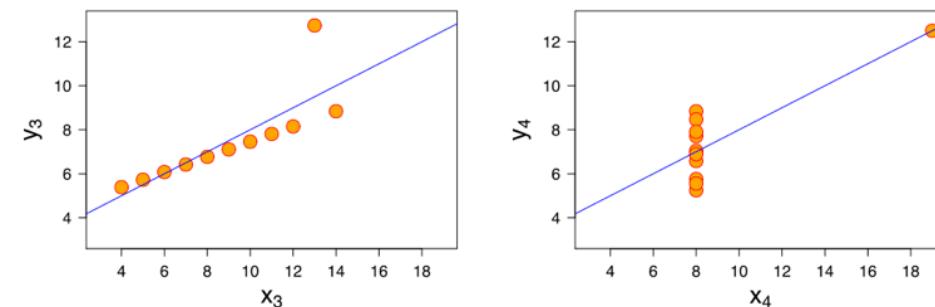
Find anomalies

- Do some basic stats:
 - Skewedness
 - Range (e.g., top mark is 10 but recorded as 100)
 - Standard deviations (measure of distance from mean, 2+ Std -> outlier)

Null rules

- How is missing data recorded (blank, NULL, unknown, -, ?, etc)

Fixing issues may require transforming data...



https://en.wikipedia.org/wiki/Standard_deviation#/media/File:Standard_deviation_diagram.svg

Wombat population survey - modified

Rows: 19548 | Columns: 8

Table  Statistics 



☐	#	RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)	
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☐	4	Row3	4	28		40	Ansons-1	2002	Modified land	0	
☐	5	Row4	5	28	40	60	Ansons-1	2002	Modified land		
☐	6	Row5	6	28	60	100	Ansons-1	2002	Modified land	0	
☐	7	Row6	7	29	0	5	Ansons-1	2003	Modified land	0	
☐	8	Row7	8	29	5	10	Ansons-1		Modified land	0	
☐	9	Row8	9	29	10	20	Ansons-1	2003	Modified land	0	
☐	10	Row9	10	29	20	40	Ansons-1	2003	Modified land	0	

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Data Transformation

- Summary of data set
 - Example;
 - daily -> weekly -> monthly -> yearly

Vehicle accidents in Victoria

	A	B	C	D	E	F	G
1	ACCIDENT_NO	ACCIDENT_C	ACCIDENT_T	ACCIDENT_TYPE	ACCIDENT_TYPE_DESC	DAY_OF_WEEK	DAY_WEEK_D
2	T20120000046	1/1/2012	16:25:00	2	Struck Pedestrian	1	Sunday
3	T20120000062	1/1/2012	18:00:00	4	Collision with a fixed obj	1	Sunday
4	T20120000060	1/1/2012	19:40:00	6	Vehicle overturned (no c	1	Sunday
5	T20120000009	1/1/2012	2:25:00	4	Collision with a fixed obj	1	Sunday
6	T20120000059	1/1/2012	18:05:00	4	Collision with a fixed obj	1	Sunday
7	T20120000058	1/1/2012	6:40:00	8	No collision and no obje	1	Sunday
8	T20120000056	1/1/2012	16:15:00	4	Collision with a fixed obj	1	Sunday
9	T20120000054	1/1/2012	18:20:00	1	Collision with vehicle	1	Sunday
10	T20120000053	1/1/2012	16:50:00	1	Collision with vehicle	1	Sunday
11	T20120000028	1/1/2012	4:00:00	4	Collision with a fixed obj	1	Sunday
12	T20120000050	1/1/2012	15:00:00	4	Collision with a fixed obj	1	Sunday
13	T20120005091	1/1/2012	15:30:00	4	Collision with a fixed obj	1	Sunday
14	T20120000602	1/1/2012	1:00:00	8	No collision and no obje	1	Sunday

Row Labels	Count of ACCIDENT_NO
2012	13912
2013	14023
2014	14503
2015	15684
2016	15779
2017	13351
2018	12763
2019	14396
2020	12170
2021	14009
2022	15016
2023	15513
Qtr1	4006
Jan	1267
Feb	1285
Mar	1454
Qtr2	3775
Qtr3	3916
Qtr4	3816
2024	7576
Grand Total	178695

Mar	1454
1/3/2023	36
2/3/2023	32
3/3/2023	44
4/3/2023	48

<https://discover.data.vic.gov.au/dataset/victoria-road-crash-data>

Data Transformation

Pivot table

- Aggregation that produces a table (grouping columns become rows)

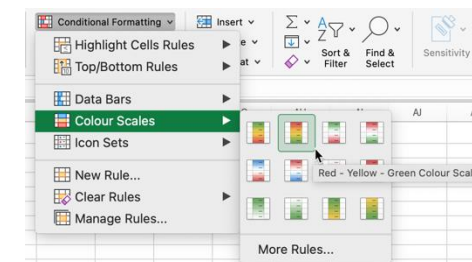
Count of ACCIDENT_NO	Colu	ibels												
Row Labels	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Grand Total
Monday	1935	1922	1927	2085	2178	1896	1777	1935	1612	1856	2039	2139	1042	24343
Tuesday	1939	2091	1994	2234	2219	1918	1908	2162	1757	1976	2172	2231	1166	25767
Wednesday	2026	2074	2194	2292	2336	2033	1890	2200	1856	2125	2158	2429	1160	26773
Thursday	2217	2119	2189	2395	2338	1954	1953	2203	1890	2135	2239	2294	1139	27065
Friday	2189	2197	2213	2528	2507	2107	1997	2352	1948	2297	2351	2447	1134	28267
Saturday	1925	1984	2120	2187	2204	1787	1745	1942	1690	1889	2172	2201	1009	24855
Sunday	1681	1636	1866	1963	1997	1656	1493	1602	1417	1731	1885	1772	926	21625
Grand Total	13912	14023	14503	15684	15779	13351	12763	14396	12170	14009	15016	15513	7576	178695

<https://discover.data.vic.gov.au/dataset/victoria-road-crash-data>

Data Transformation

Pivot table

- Aggregation that produces a table (grouping columns become rows)



Conditional formatting:
How to make quick heat map in Excel

Count of ACCIDENT_NO	Colu	Labels													
Row Labels	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Grand Total	
Monday	1935	1922	1927	2085	2178	1896	1777	1935	1612	1856	2039	2139	1042	24343	
Tuesday	1939	2091	1994	2234	2219	1918	1908	2162	1757	1976	2172	2231	1166	25767	
Wednesday	2026	2074	2194	2292	2336	2033	1890	2200	1856	2125	2158	2429	1160	26773	
Thursday	2217	2119	2189	2395	2338	1954	1953	2203	1890	2135	2239	2294	1139	27065	
Friday	2189	2197	2213	2528	2507	2107	1997	2352	1948	2297	2351	2447	1134	28267	
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<https://discover.data.vic.gov.au/dataset/victoria-road-crash-data>

Data Transformation

Pivot table

- Aggregation that produces a table (grouping columns become rows)

For differences in magnitude use **saturation** (intensity/purity of colour)

Count of ACCIDENT_NO	Colu	Labels												
Row Labels	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Grand Total
Monday	1935	1922	1927	2085	2178	1896	1777	1935	1612	1856	2039	2139	1042	24343
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Thursday	2217	2119	2189	2395	2338	1954	1953	2203	1890	2135	2239	2294	1139	27065
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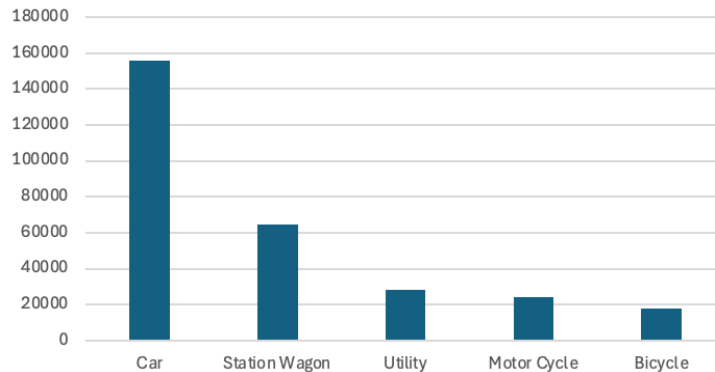
Conditional formatting in excel is relative
Apply separately to columns and totals

<https://discover.data.vic.gov.au/dataset/victoria-road-crash-data>

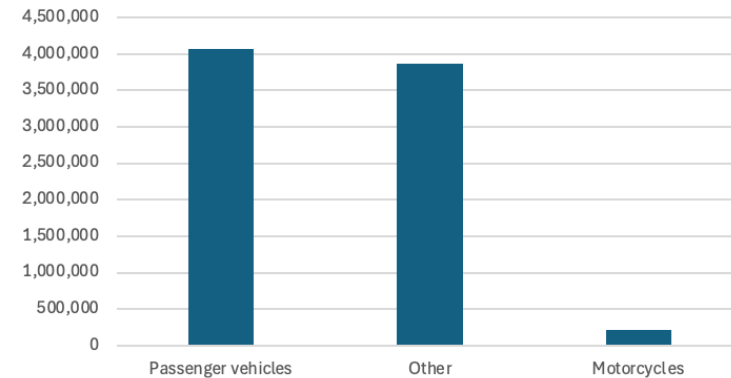
Data Transformation

Attribute construction

- Creating new attributes
 - Example;
 - Vehicle accidents -> convert to...
vehicle accidents per 100,000
vehicle registrations



Vehicle accidents in Victoria



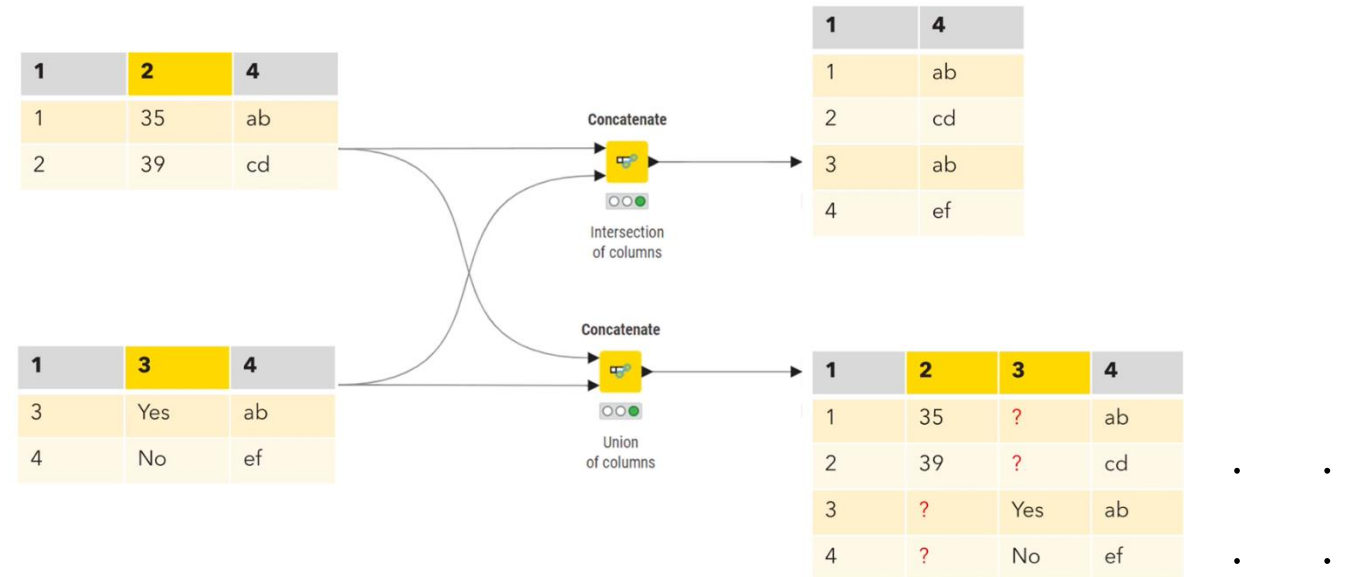
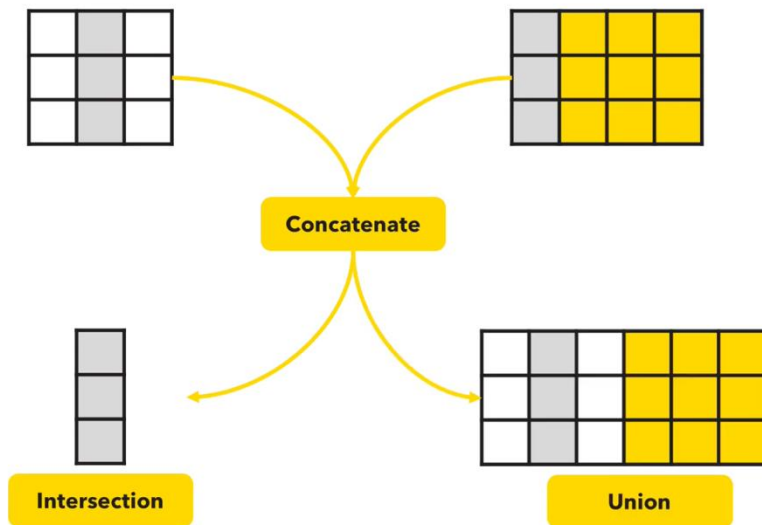
Vehicle registrations in Victoria 2023

<https://www.bitre.gov.au/publications/2024/road-vehicles-australia-january-2024>

Data Transformation - Combining data sets

Integration

- Concatenate (aka merge, union, combine)



<https://www.youtube.com/watch?v=FiEiKUfNN0Q>

Data Transformation - Combining data sets

Integration

- Joins
 - joins rows from two data tables based on common values (e.g., ID) in one or more columns

Left table

Customer ID	Gender	Age
11000	Male	44
11001	Female	45
11002	Male	45

Inner Join



Right table

Customer ID	Product
11000	CO Investment
11001	Fund Manager+
11004	Private Investment

Joined table

Customer ID	Gender	Age	Product
11000	Male	44	CO Investment
11001	Female	45	Fund Manager+

Left table

Customer ID	Gender	Age
11000	Male	44
11001	Female	45
11002	Male	45

Left Outer Join



Right table

Customer ID	Product
11000	CO Investment
11001	Fund Manager+
11004	Private Investment

Joined table

Customer ID	Gender	Age	Product
11000	Male	44	CO Investment
11001	Female	45	Fund Manager+
11002	Male	45	?

Left table

Customer ID	Gender	Age
11000	Male	44
11001	Female	45
11002	Male	45

Right Outer Join



Right table

Customer ID	Product
11000	CO Investment
11001	Fund Manager+
11004	Private Investment

Joined table

Customer ID	Gender	Age	Product
11000	Male	44	CO Investment
11001	Female	45	Fund Manager+
11004	?	?	Private Investment

<https://www.youtube.com/watch?v=1Rqb1jxb86o>

Class Exercise 1.0 Data Processing with KNIME

Scenario

Imagine you have been asked to create a set of visualisations that allows users to explore the energy consumption of televisions currently available on the Australian market.

Overview

1. Find the data set
2. Open data set in KNIME
3. Become familiar with data set
4. Clean the data set
5. Transform the data set

